Ariel Ooms
Honors Option - World Average Years of Formal Schooling
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Data and Research Questions

I am investigating how the average years of schooling varies by country around the world. My data came from Our World in Data and can be found at this website: https://ourworldindata.org/grapher/mean-years-of-schooling-long-run?tab=table. The variables available are country, year, and average years of formal education for individuals between ages 15-64.The data covers years from 1870 through 2020.

My primary questions are which countries saw the largest change in average years of schooling between 2000-2020 and which countries saw the least amount of change, is the average years of schooling in the United States between 2000-2020 statistically different from Canada, and how did the average years schooling change in the 21st century worldwide?

Methods and Analysis

I began by cleaning the data by changing the column names to more understandable names. This changed Combined.average.years.of.education.for.15.64.years.male.and.female.youth.and.adults to MeanSchooling. I first ran a general summary of the dataset to observe key values such as the mean, median, and quartiles of the average years of schooling (figure 1). I also filtered my data first into years 2000 and 2020 so I could later calculate the change in schooling between those two years. I filtered the original dataset into two datasets, one for the United States and one for Canada.

I first looked at which countries saw the largest and smallest change in average years of formal schooling between 2000 and 2020. I subtracted the averages from 2000 from 2020 and sorted by largest change. I used bar plots to show the top 5 countries with the biggest change in average schooling and the top 5 countries with the smallest change in average schooling.

To answer my next research question, is the average years of schooling in the United States across the entire dataset timeframe statistically different from Canada, I created a null and alternate hypothesis. The null hypothesis is that there is no difference between the average years of schooling in the United States and Canada between 1870-2020 ($\mu_{US} = \mu_{Canada}$). My alternate hypothesis is that the two averages are different ($\mu_{US} \neq \mu_{Canada}$). To test this hypothesis, I used a t-test to compare the United States and Canada. This t-test will determine if the means of the two countries are statistically different and represents an actual difference in the two populations.

I utilized maps to visually answer my third research question, how did the average years schooling change in the 21st century worldwide. I used the R libraries ggplot and rnaturalearth to create global maps and color the countries by average schooling.

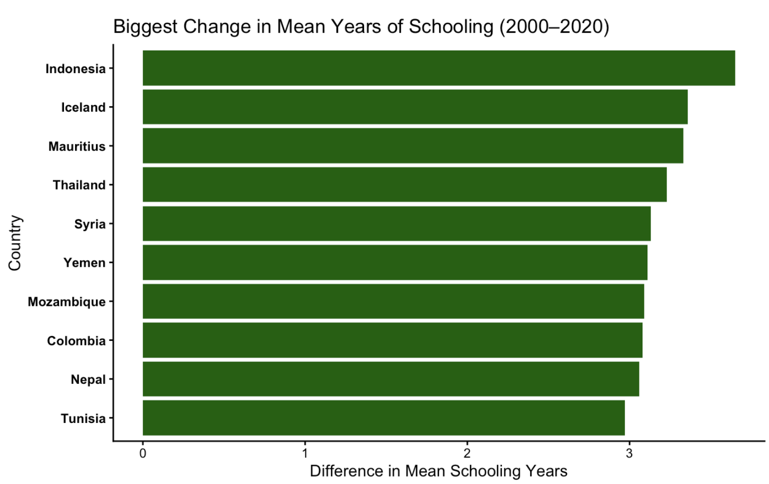
Results

Figure 1 Summary of the Mean Schooling variable within the dataset

```
MeanSchooling
Min.   : 0.000
1st Qu.: 0.540
Median : 2.760
Mean   : 3.813
3rd Qu.: 6.370
Max.   :13.740
```

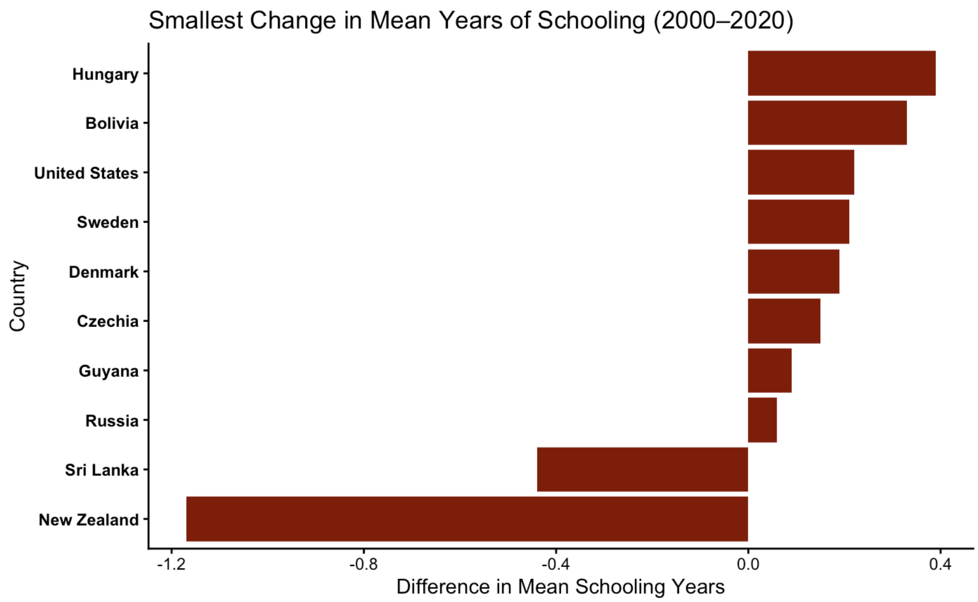
The summary of my data shows that there are countries with either no data or an average of zero years of formal schooling. The maximum average years of formal schooling was 13.740 and the average is 3.813 years of schooling. 75% of the countries in the dataset had an average of 3.813 years of formal schooling. Since the median is lower than the mean, 2.760 compared to 3.813, this indicates that there are a few countries with much higher averages that are pulling the total average up.

Figure 2a Top 10 Countries with the Largest Change in Mean Schooling 2000-2020



The 10 countries that saw the greatest increase in average years of schooling between 2000 and 2020 were Indonesia, Iceland, Mauritius, Thailand, Syria, Yemen, Mozambique, Colombia, Nepal, and Tunisia. Indonesia saw the greatest increase of about 3.5 additional years, while the remaining nine countries also saw an increase of about 3 years.

Figure 2b Top 10 Countries with the Smallest Change in Mean Schooling 2000-2020



The top 10 countries that saw the smallest change in average years of schooling between 2000-2020 were Hungary, Bolivia, United States, Sweden, Denmark, Czechia, Guyana, Russia, Sri Lanka, and New Zealand. Only Sri Lanka and New Zealand saw a decrease in the average years of schooling. The other eight countries saw just a very slight increase in the average number of years in formal schooling all ranging between 0 and 0.4. One thing to consider is if the countries already had a very high average, this decrease may not indicate poor schooling, but instead a slight decrease in an already large number of average years.

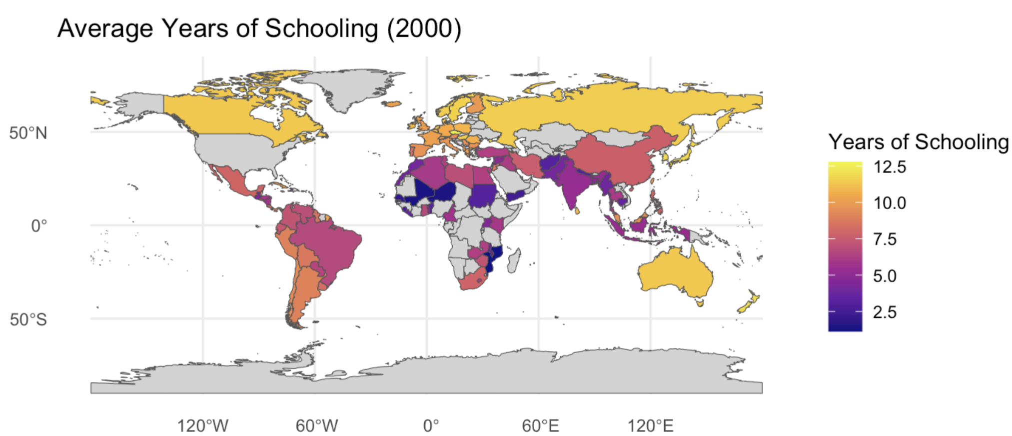
Figure 3 Welch Two Sample t-test of Mean Schooling in Canada and the United States

```
        Welch Two Sample t-test

data:  schooling_can$MeanSchooling and schooling_US$MeanSchooling
t = -1.9463, df = 59.254, p-value = 0.05637
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -3.21799677  0.04444838
sample estimates:
mean of x mean of y 
 7.527419  9.114194 
```

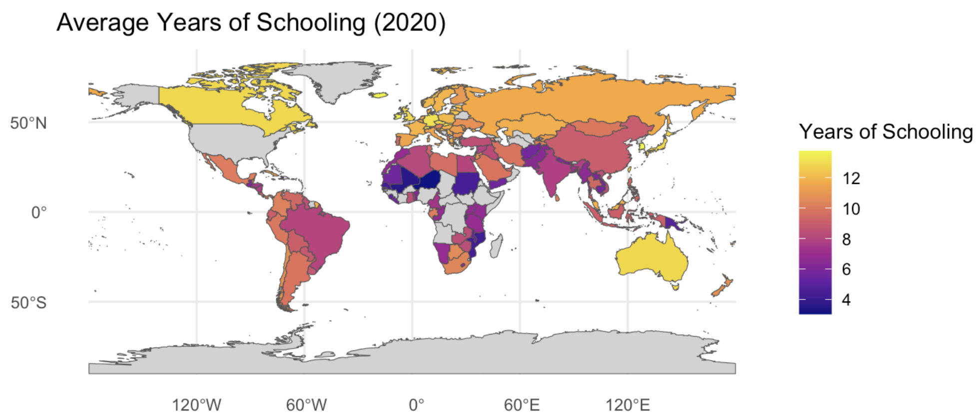
The results from the t-test show that the difference in average years of formal schooling in the United States and Canada might be attributable to random chance and therefore the null hypothesis cannot be rejected. This is because the p-value is 0.05637 and is not below the level of significance 0.05. This is also because the 95% confidence interval is from -3.22 to 0.044 and covers the null value, indicating that there is no significant difference in the two countries' average schooling.

Figure 4a Global Map of Average Years of Schooling 2000



This map shows that Russia, Canada, and Australia have the highest averages in formal schooling. Several countries in Africa appear to have the lowest averages. Europe, Asia, and South America all have a range of averages toward the median years of schooling.

Figure 4b Global Map of Average Years of Schooling 2020



This map shows that Canada, Russia, and Australia all maintained their highest level of formal schooling and there appears to be very little change overall in the world in average years of formal schooling. Some countries in South America appear to have decreased in average years of schooling, while some countries in Northern Africa appear to have increased slightly.

Communicating Science

Being able to communicate scientific findings in a way that the general audience member will understand is a necessary skill for anyone pursuing a career in research or science. This research has identified the spread of the average years of schooling across the globe as well as pinpointed countries that are doing well with expanding their formal schooling and countries that need to put more attention toward their education systems to find any reasons why their mean years of schooling is decreasing. The statistical test used to determine if there was a difference in the average years of schooling in Canada and the United States indicated that there was no significant difference between the countries. This project allowed me to provide suggestions for which countries are doing well and which need to improve in their education. It also allowed me to conclude that the United States and Canada do not have a difference in their average years of formal education.